**§ Groups** A **group** is a nonempty set $G$ under the (binary) operation $* : S \times S \to S$, denoted by $(G, *)$, such that

- **Closure:** For all $a, b \in G$, $a * b \in G$.
- **Associativity:** For all $a, b, c \in G$, $(a * b) * c = a * (b * c)$.
- **Existence of Identity:** There exists an element $e \in G$ such that for all $a \in G$, $e * a = a * e = a$.
- **Existence of Inverse:** For each $a \in G$, there exists an element $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$.

1. Show that $(\mathbb{Z}, +)$ is a group.

**Proof**: Let $G = \mathbb{Z}$ and let the operation be $+$. We will show that $(G, +)$ satisfies all four group properties.

- **Closure:** For any $a, b \in G$, $a + b \in G$ since the sum of two integers is an integer.
- **Associativity:** For any $a, b, c \in G$, $(a + b) + c = a + (b + c)$ by the associativity of integer addition.
- **Existence of Identity:** The identity element is 0 since for any $a \in G$, $0 + a = a + 0 = a$.
- **Existence of Inverse:** For each $a \in G$, the inverse element is $-a$ since $a + (-a) = (-a) + a = 0$.

Since all four group properties are satisfied, it follows that $(\mathbb{Z}, +)$ is a group.

■

2. Show that $(\mathbb{Z}, \cdot)$ is not a group.

**Proof**: Let $G = \mathbb{Z}$ and let the operation be $\cdot$. We will show that $(G, \cdot)$ does not satisfy all four group properties.

- **Closure:** For any $a, b \in G$, $a \cdot b \in G$ since the product of two integers is an integer.
- **Associativity:** For any $a, b, c \in G$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ by the associativity of integer multiplication.
- **Existence of Identity:** The identity element is 1 since for any $a \in G$, $1 \cdot a = a \cdot 1 = a$.
- **Existence of Inverse:** For each $a \in G$, there does not exist an inverse element $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = 1$ unless $a = 1$ or $a = -1$. For example, if $a = 2$, then there is no integer $b$ such that $2 \cdot b = 1$.

Since the existence of inverse property is not satisfied for all elements in $G$, it follows that $(\mathbb{Z}, \cdot)$ is not a group.

■

3. Show that $(\mathbb{Q}, +)$ is a group.

**Proof**: Let $G = \mathbb{Q}$ and let the operation be $+$. We will show that $(G, +)$ satisfies all four group properties.

- **Closure:** For any $a, b \in G$, $a + b \in G$ since the sum of two rational numbers is a rational number.
- **Associativity:** For any $a, b, c \in G$, $(a + b) + c = a + (b + c)$ by the associativity of rational number addition.
- **Existence of Identity:** The identity element is 0 since for any $a \in G$, $0 + a = a + 0 = a$.
- **Existence of Inverse:** For each $a \in G$, the inverse element is $-a$ since $a + (-a) = (-a) + a = 0$.

Since all four group properties are satisfied, it follows that $(\mathbb{Q}, +)$ is a group.

■

4. Show that $(\mathbb{Q}, \cdot)$ is not a group.

**Proof**: Let $G = \mathbb{Q}$ and let the operation be $\cdot$. We will show that $(G, \cdot)$ does not satisfy all four group properties.

- **Closure:** For any $a, b \in G$, $a \cdot b \in G$ since the product of two rational numbers is a rational number.
- **Associativity:** For any $a, b, c \in G$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ by the associativity of rational number multiplication.
- **Existence of Identity:** The identity element is 1 since for any $a \in G$, $1 \cdot a = a \cdot 1 = a$.

- **Existence of Inverse:** For each $a \in G$, there does not exist an inverse element $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = 1$ if $a = 0$. For example, if $a = 0$, then there is no rational number $b$ such that $0 \cdot b = 1$.

Since the existence of inverse property is not satisfied for all elements in $G$, it follows that $(\mathbb{Q}, \cdot)$ is not a group.

■

5. Show that $(\mathbb{Q}^*, \cdot)$ is a group, where $\mathbb{Q}^* = \{q \in \mathbb{Q} \mid q \neq 0\}$.

**Proof:** Let $G = \mathbb{Q}^*$ and let the operation be $\cdot$. We will show that $(G, \cdot)$ satisfies all four group properties.

- **Closure:** For any $a, b \in G$, $a \cdot b \in G$ since the product of two nonzero rational numbers is a nonzero rational number.
- **Associativity:** For any $a, b, c \in G$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ by the associativity of rational number multiplication.
- **Existence of Identity:** The identity element is 1 since for any $a \in G$, $1 \cdot a = a \cdot 1 = a$.
- **Existence of Inverse:** For each $a \in G$, the inverse element is $a^{-1} = \frac{1}{a}$ since $a \cdot a^{-1} = a^{-1} \cdot a = 1$.

Since all four group properties are satisfied, it follows that $(\mathbb{Q}^*, \cdot)$ is a group.

■

6. Show that $(\mathbb{R}^+, +)$ is a group.

**Proof:** Let $G = \mathbb{R}^+$ and let the operation be $+$. We will show that $(G, +)$ does not satisfy all four group properties.

- **Closure:** For any $a, b \in G$, $a + b \in G$ since the sum of two positive real numbers is a positive real number.
- **Associativity:** For any $a, b, c \in G$, $(a + b) + c = a + (b + c)$ by the associativity of real number addition.
- **Existence of Identity:** There does not exist an identity element $e \in G$ such that for all $a \in G$, $e + a = a + e = a$ since the only candidate for the identity element is 0, which is not in $G$.
- **Existence of Inverse:** For each $a \in G$, there does not exist an inverse element $a^{-1} \in G$ such that $a + a^{-1} = a^{-1} + a = e$ since the only candidate for the inverse element is $-a$, which is not in $G$.

Since the existence of identity and existence of inverse properties are not satisfied, it follows that $(\mathbb{R}^+, +)$ is not a group.

■

7. Show that $(\mathbb{R}^+, \cdot)$ is a group.

**Proof:** Let $G = \mathbb{R}^+$ and let the operation be $\cdot$. We will show that $(G, \cdot)$ satisfies all four group properties.

- **Closure:** For any $a, b \in G$, $a \cdot b \in G$ since the product of two positive real numbers is a positive real number.
- **Associativity:** For any $a, b, c \in G$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ by the associativity of real number multiplication.
- **Existence of Identity:** The identity element is 1 since for any $a \in G$, $1 \cdot a = a \cdot 1 = a$.
- **Existence of Inverse:** For each $a \in G$, the inverse element is $a^{-1} = \frac{1}{a}$ since $a \cdot a^{-1} = a^{-1} \cdot a = 1$.

Since all four group properties are satisfied, it follows that $(\mathbb{R}^+, \cdot)$ is a group.

■

**§ Abelian Groups** A group $(G, *)$ is called an **Abelian group** (or **commutative group**) if (and only if) **commutative**, that is,

For all $a, b \in G$, $a * b = b * a$.

1. Let $G$ be a nonempty set of $2 \times 2$ matrices. Then, $(G, \cdot)$ is not an Abelian group, where $\cdot$ is the matrix multiplication operation.

**Proof**: Let $G$ be the set of all $2 \times 2$ matrices with real entries, and let the operation be matrix multiplication $\cdot$. We will show that $(G, \cdot)$ does not satisfy the commutative property. Consider the following two matrices in $G$:

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}.$$

Now, we compute $A \cdot B$ and $B \cdot A$:

$$A \cdot B = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} 1+6 & 2+2 \\ 0+3 & 0+1 \end{pmatrix} = \begin{pmatrix} 7 & 4 \\ 3 & 1 \end{pmatrix},$$

and

$$B \cdot A = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1+0 & 2+0 \\ 3+0 & 6+1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}.$$

Since $A \cdot B = \begin{pmatrix} 7 & 4 \\ 3 & 1 \end{pmatrix}$ is not equal to $B \cdot A = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}$. Therefore, $(G, \cdot)$ does not satisfy the commutative property, and hence it is not an Abelian group.

$\blacksquare$

**§ Finite Groups** A group $(G, *)$ is called a **finite group** if (and only if) the order of $G$, denoted by $|G|$, is finite; that is,

$$|G| < \infty.$$

**§ Group** A nonempty set $G$ with a binary operation $*: G \times G \to G$, written $(G, *)$, is a **group** if and only if

- **Closure:** For all $a, b \in G$, $a * b \in G$.
- **Associativity:** For all $a, b, c \in G$, $(a * b) * c = a * (b * c)$.
- **Identity:** There exists $e \in G$ such that for all $a \in G$, $e * a = a * e = a$.
- **Inverses:** For each $a \in G$ there exists $b \in G$ such that $a * b = b * a = e$.

Groups are generalizations of the familiar products of addition and of multiplication of real numbers. Remember, though, that a set with a product is a group if and only if the four axioms are satisfied!

For instance, $\mathbb{Z}$ is a group under addition but not under multiplication (for more than one reason). The set $\mathbb{Q}$ is also group under addition but not under multiplication. The set $\mathbb{R}^+$of positive real numbers is a group under multiplication but not under addition.

The groups above are infinite, but many interesting groups are finite. The order of a group $G$ is the number of elements in $G$, denoted $|\boldsymbol{G}|$. If (and only if) $|\boldsymbol{G}| < \infty$, then $G$ is a finite group.

The smallest group consists of a single (identity) element, usually denoted $e$ (or sometimes $i$ when the group elements are functions). Here's an example of a group of order 2: Claim: The set $\{-1, 1\}$ under multiplication is a group. Proof: Consider the set $\{-1, 1\}$ under multiplication, denoted $\cdot$. Since $-1 \cdot -1 = 1 = 1 \cdot 1$ and $-1 \cdot 1 = -1 = 1 \cdot -1$, then $\{-1, 1\}$ is closed under multiplication. Since multiplication of real number is associative, then multiplication on $\{-1, 1\}$ is associative. Since $-1 \cdot 1 = 1 \cdot -1 = -1$ and $1 \cdot 1 = 1$, then 1 is the identity under multiplication. Since $-1 \cdot -1 = 1$ and $1 \cdot 1 = 1$, then both -1 and 1 have multiplicative inverses. Since all four axioms are satisfied, $(\{-1, 1\}, \cdot)$ is a group. QED

When the order of a group is small, it is often very useful to write out the product table for the group.

For instance, let $G$ be a group of order two with product $*$. Then $G = \{e, a\}$ where $e$ is the identity and $a$ denotes the single non-identity element. The product table for $G$ is:

| $*$ | $e$ | $a$ |
|---|---|---|
| $e$ | $e$ | $a$ |
| $a$ | $a$ | $e$ |

This is the "only" group of order 2. By which we mean that all groups of order 2 must be isomorphic to this group.

**§ Group Isomorphism** Let $(G, *)$ and $(G', \star)$ be groups. An **isomorphism** is a bijection $\phi: G \to G'$ such that $\phi(a * b) = \phi(a) \star \phi(b)$ for all $a, b \in G$. We write $G \simeq G'$ if such a map exists.

Claim: The group $(\{-1, 1\}, \cdot)$ is isomorphic to the group $(G, *)$ with product table above. Proof: Let $\phi: \{-1, 1\} \to \{e, a\}$ be the function given by $\phi(1) = e$ and $\phi(-1) = a$. Then by inspection $\phi$ is one-to-one and onto, and hence a bijection. Also, $\phi(1 \cdot 1) = \phi(1) = e = e * e = \phi(1) * \phi(1)$, $\phi(1 \cdot -1) = \phi(-1) = a = e * a = \phi(1) * \phi(-1)$, $\phi(-1 \cdot 1) = \phi(-1) = a = a * e = \phi(-1) * \phi(1)$, and $\phi(-1 \cdot -1) = \phi(1) = e = a * a = \phi(-1) * \phi(-1)$. Hence, $\phi$ preserve operations, and $(\{-1, 1\}, \cdot) \simeq (\{e, a\}, *)$. QED Can you think of another group of order 2 ? What about the set $\{[0]_2, [1]_2\}$ of congruence classes modulo two under addition? Can you prove this is isomorphic to $(G, *)$ with product table above?

Fill in the product tables below. Let $e$ denote the identity element. Hint: How many times can an element appear in a given row (or column)? How many times must an element appear in a given row (or column)? If $g * h = h$ in a group $(G, *)$ with $g, h \in G$, then what is known about $g$ ?

Up to isomorphism, there is one group of order 3. You should understand why after filling in this product table.

| | $e$ | $a$ | $b$ |
|---|---|---|---|
| $e$ | | | |
| $a$ | | | |
| $b$ | | | |

Up to isomorphism, there are two groups of order 4. Identify the first one by completing this product table.

| | $e$ | $a$ | $b$ | $c$ |
|---|---|---|---|---|
| $e$ | | | | |
| $a$ | | $b$ | | |
| $b$ | | | | |
| $c$ | | | | |

Identify the other group of order 4 which is not isomorphic to the previous one. Hint: confirm your answer is not isomorphic to the previous group of order 4 under bijections like $b \leftrightarrow c$ or $a \leftrightarrow b$.

| | $e$ | $a$ | $b$ | $c$ |
|---|---|---|---|---|
| $e$ | | | | |
| $a$ | | | | |
| $b$ | | | | |
| $c$ | | | | |
| $c$ | | | | |

Recall Wbwk8, Problem 4. Fix $n \geq 2$ and consider $\mathbb{Z}_n$, the set of equivalence classes under the relation congruence modulo $n$ on $\mathbb{Z}$. Are the following propositions true or false?

1. Since addition on $\mathbb{Z}$ is associative, so is addition on $\mathbb{Z}_n$.
2. Since multiplication on $\mathbb{Z}$ is commutative, so is multiplication on $\mathbb{Z}_n$.
3. For all $a, b, c, d \in \mathbb{Z}$, if $[a] = [b]$ and $[c] = [d]$, then $[ac] = [bd]$.
4. Multiplication is well-defined on $\mathbb{Z}_n$.
5. Since multiplication on $\mathbb{Z}$ is associative, so is multiplication on $\mathbb{Z}_n$.
6. Addition is well-defined on $\mathbb{Z}_n$.
7. For all $a, b, c, d \in \mathbb{Z}$, if $[a] = [b]$ and $[c] = [d]$, then $[a + c] = [b + d]$.
8. Since addition on $\mathbb{Z}$ is commutative, so is addition on $\mathbb{Z}_n$.

Now, compute the following sum and product tables for $n = 6$.

| $+$ | [0] | [1] | [2] | [3] | [4] | [5] |
|---|---|---|---|---|---|---|
| [0] | | | | | | |
| [1] | | | | | | |
| [2] | | | | | | |
| [3] | | | | | | |
| [4] | | | | | | |
| [5] | | | | | | |

| $\cdot$ | [0] | [1] | [2] | [3] | [4] | [5] |
|---|---|---|---|---|---|---|
| [0] | | | | | | |
| [1] | | | | | | |
| [2] | | | | | | |
| [3] | | | | | | |
| [4] | | | | | | |
| [5] | | | | | | |

Fix $n \geq 2$ and consider $\mathbb{Z}_n$, the set of equivalence classes under the relation congruence modulo $n$ on $\mathbb{Z}$. By definition, $\mathbb{Z}_n^* = \mathbb{Z}_n \backslash \{[0]\}$. Are the following propositions true or false?

(**false**) Under multiplication $[0]$ is the identity element in $\mathbb{Z}_n$.

(**false**) The set $\mathbb{Z}_n$ is a group under the product of multiplication.

(**true**) There exists an identity element in $\mathbb{Z}_n$ under addition.

(**false**) The set $\mathbb{Z}_n^*$ is a group under the product of multiplication.

(**false**) For every $[a] \in \mathbb{Z}_n$ there exists a $[b] \in \mathbb{Z}_n$ such that $[ab]$ is the multiplicative identity.

(**true**) Under multiplication $[1]$ is the identity element in $\mathbb{Z}_n^*$.

(**false**) Under multiplication $[0]$ is the identity element in $\mathbb{Z}_n^*$ when $n$ is prime.

(**true**) Under addition $[0]$ is the identity element in $\mathbb{Z}_n$.

(**true**) There exist inverses in $\mathbb{Z}_n^*$ under multiplication when $n$ is prime.

(**true**) There exists an identity element in $\mathbb{Z}_n^*$ under multiplication.

(**true**) The set $\mathbb{Z}_n^*$ is a group under the product of multiplication when $n$ is prime.

(**false**) For every $[a] \in \mathbb{Z}_n^*$ there exists a $[b] \in \mathbb{Z}_n^*$ such that $[ab]$ is the multiplicative identity.

(**true**) For every $[a] \in \mathbb{Z}_n$ there exists a $[b] \in \mathbb{Z}_n$ such that $[a+b]$ is the additive identity.

(**true**) Under multiplication $[1]$ is the identity element in $\mathbb{Z}_n^*$ when $n$ is prime.

(**true**) There exists an identity element in $\mathbb{Z}_n$ under multiplication.

(**false**) There exist inverses in $\mathbb{Z}_n$ under multiplication.

(**false**) Under multiplication $[0]$ is the identity element in $\mathbb{Z}_n^*$.

(**true**) For every $[a] \in \mathbb{Z}_n^*$ there exists a $[b] \in \mathbb{Z}_n^*$ such that $[ab]$ is the multiplicative identity when $n$ is prime.

(**false**) Under addition $[1]$ is the identity element in $\mathbb{Z}_n$.

(**true**) The set $\mathbb{Z}_n$ is a group under the product of addition.

(**true**) Under multiplication $[1]$ is the identity element in $\mathbb{Z}_n$.

(**true**) There exists an identity element in $\mathbb{Z}_n^*$ under multiplication when $n$ is prime.

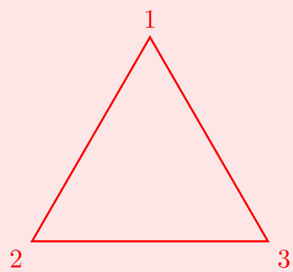
(**true**) There exist inverses in $\mathbb{Z}_n$ under addition.

(**false**) There exist inverses in $\mathbb{Z}_n^*$ under multiplication.

**§ Nonabelian Group: Symmetries of an Equilateral Triangle** A group $(G, *)$ is **abelian** if and only if $a * b = b * a$ for all $a, b \in G$. While many familiar groups are abelian, most groups are not! Hence, it is very important to understand well some examples of nonabelian groups.

The smallest nonabelian group has order 6. One way to understand this group is as the symmetries of an equilateral triangle.

Consider an equilateral triangle with vertex labels 1, 2, and 3:




Let $f$ be a reflection through the vertical axis. So $f$ fixes point 1 and switches 2 and 3.

Let $g$ be a rotation 120° clockwise around the center point. So $g$ sends 1 to 2, 2 to 3, and 3 to 1.

There are four other symmetries of an equilateral triangle. What are they in terms of either an axis of reflection or an angle of rotation?

The composition of two symmetries is another symmetry, and this product is associative. There is an identity symmetry, and every symmetry has an inverse. Why? How would you justify these statements?

Hence, the symmetries of an equilateral triangle form a group under composition.

Match each symmetry on the left with an equivalent one from the right if possible. The notation $fg$ means $f \circ g$, so first rotate 120°, then reflect through the vertical axis.

A. $gf$

B. $g^{-1}$

C. $g^2 f$

D. $f$

E. $f^2$

F. None of the above

Match the following.

1. $f^{-1}$
2. $g^3$
3. $fg^2$
4. $fg$
5. $g^2$

Fill in the product table below using the given labels for the symmetries. Let $e$ denote the identity symmetry. Enter $g^2$ as "gg" and $fg^2$ as "fgg".

| | $e$ | $f$ | $g$ | $g^2$ | $fg$ | $fg^2$ |
|---|---|---|---|---|---|---|
| $e$ | | | | | | |
| $f$ | | | | | | |
| $g$ | | | | | | |
| $g^2$ | | | | | | |
| $fg$ | | | | | | |
| $fg^2$ | | | | | | |

**§ Order of an Element** Let $(G, *)$ be a group with identity $e$. The **order** of an element $a \in G$, denoted $o(a)$, is the least positive integer $m$ such that $a^m = e$. If no such $m$ exists, we say $a$ has **infinite order**.

Let $(G, *)$ be the group of symmetries of an equilateral triangle under composition. What are the truth values of the following propositions?

1. Every element in $(G, *)$ has order 2 or 3.
2. $fg = g^{-1}f$ but $g \neq g^{-1}$.
3. The order of $(G, *)$ is 6.
4. The order of $g^2$ is 6.
5. The order of $f$ is 2.
6. $(fg)^2 \neq f^2g^2$.
7. For all elements $h \in G$, $h^6 = e$.
8. The order of $g$ is 3.
9. The order of every element in $(G, *)$ divides the order of the group.
10. $g^2(fg) = (fg)g^2$.
11. $(G, *)$ is an abelian group.

Up to isomorphism, there is a unique group of order 5 . Let $G = \{a, b, c, d, e\}$ be a group with product *. Complete the product table for $(G, *)$ below.

| $*$ | $e$ | $a$ | $b$ | $c$ | $d$ |
|---|---|---|---|---|---|
| $e$ | $e$ | $a$ | $b$ | $c$ | $d$ |
| $a$ | $a$ | $b$ | $d$ | $e$ | $c$ |
| $b$ | $b$ | $d$ | $c$ | $a$ | $e$ |
| $c$ | $c$ | $e$ | $a$ | $d$ | $b$ |
| $d$ | $d$ | $c$ | $e$ | $b$ | $a$ |

The congruence classes modulo 5 are also a group of order 5 under addition. Show that $(\mathbb{Z}_5, +)$ is isomorphic to $(G, *)$ in several different ways by completing the various bijections $\phi : \mathbb{Z}_5 \to G$ below.

Match the elements of $\mathbb{Z}_5$ on the left with their images in $G$. Remember that the bijection $\phi$ must preserve operations to be an isomorphism.

1. Suppose $\phi$ maps [1] to $a$.

   **Answer**:

   - $\phi([0]) \to e$
   - $\phi([1]) \to a$
   - $\phi([2]) \to a^2 = b$
   - $\phi([3]) \to a^3 = d$
   - $\phi([4]) \to a^4 = c$

   □

2. Suppose $\phi$ maps [1] to $b$.

   **Answer**:

   - $\phi([0]) \to e$
   - $\phi([1]) \to b$
   - $\phi([2]) \to b^2 = c$
   - $\phi([3]) \to b^3 = a$
   - $\phi([4]) \to b^4 = d$

   □

3. Suppose $\phi$ maps [1] to $c$.

   **Answer**:

   - $\phi([0]) \to e$
   - $\phi([1]) \to c$
   - $\phi([2]) \to c^2 = d$
   - $\phi([3]) \to c^3 = b$
   - $\phi([4]) \to c^4 = a$

   □

4. Suppose $\phi$ maps [1] to $d$.

   **Answer**:

- $\phi([0]) \to e$ (identity: $d^0 = e$)
- $\phi([1]) \to d$ ($d^1 = d$)
- $\phi([2]) \to d^2 = a$ ($d^2 = a$ from the table)
- $\phi([3]) \to d^3 = c$ ($d^3 = c$ from the table)
- $\phi([4]) \to d^4 = b$ ($d^4 = b$ from the table)

$\square$

5. Suppose $\phi$ maps [2] to $a$.

**Answer**:

- $\phi([0]) \to e$ (identity: $a^0 = e$)
- $\phi([1]) \to d$ ($a^1 = a$, but $a = d^2$, so $d$ is the element after $e$ in the cycle)
- $\phi([2]) \to a$ (by assumption)
- $\phi([3]) \to c$ ($a^3 = c$ from the table)
- $\phi([4]) \to b$ ($a^4 = b$ from the table)

$\square$

6. Suppose $\phi$ maps [3] to $a$.

**Answer**:

- $\phi([0]) \to e$ (identity: $a^0 = e$)
- $\phi([1]) \to b$ ($a^1 = a$, $a^2 = b$)
- $\phi([2]) \to c$ ($a^3 = c$)
- $\phi([3]) \to a$ (by assumption)
- $\phi([4]) \to d$ ($a^4 = d$)

$\square$

7. Suppose $\phi$ maps [4] to $a$.

**Answer**:

- $\phi([0]) \to e$
- $\phi([1]) \to c$
- $\phi([2]) \to d$
- $\phi([3]) \to b$
- $\phi([4]) \to a$

$\square$